Appendix: Artifact Description

Artifact Description (AD)

Paper ID: `pap142`
Requested SC26 badge review: all three artifact badges.

I. Contributions and Artifacts

A. Main Contributions

- `C1`: Target Dependency Packets (TDPs) expose target-centric temporal dependencies and branch-level parallelism.
- `C2`: DDTC constructs TDPs on demand to reduce target-irrelevant work and data movement.
- `C3`: OATS and PHLE reuse overlapping dependency packets while preserving ordered target updates.
- `C4`: TempGNN is evaluated on an Alveo U280 across six temporal datasets and four TGNN models, including accelerator-baseline comparisons.

B. Computational Artifacts

TABLE I
Artifact-to-result map.

ID	Contents	Contributions and results
`A1`	Four U280 sources, xclbins, hosts, inputs, tests, and measurement workflow	`C1–C4`; U280 latency behavior used in the accelerator comparison
`A2`	`results/result.csv` and the CSV/SVG generator	`C1–C4`; Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14

Artifact: 10.5281/zenodo.21417187; source: TempGNN-Program/TempGNN; tag: `sc26-ae-pap142-v1.0`.

II. Download

```
git clone https://github.com/TempGNN-Program/
  ↪  TempGNN.git
cd TempGNN
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

III. Environment and Inputs

```
Board: AMD/Xilinx Alveo U280,
  ↪  xcu280-fsvh2892-2L-e
Platform: xilinx_u280_gen3x16_xdma_1_202211_1
OS: Ubuntu 22.04 x86_64
XRT: 2.16.204
Vitis/Vivado: 2023.2 (xclbin rebuild only)
GCC: 11.4.0
GNU Make: 4.3
Python: 3.10 or newer
```

XRT is available from https://github.com/Xilinx/XRT; Python is available from https://www.python.org/. A reviewer account on the matching U280 host is provided through the private SC26 submission channel.

`A1` uses bundled prefixes of WK, MC, RT, LM, WT, and GT with JODIE, TGN, TGAT, and APAN. Input source metadata and hashes are under `external/u280_dataset_samples/`. `A2` uses `results/result.csv`.

IV. Run on U280

`A1` exercises the TempGNN mechanisms and U280 evaluation supporting `C1–C4`.

```
source /opt/xilinx/xrt/setup.sh
make ae-core-u280 U280_CORE_DEVICE=0
  ↪  U280_CORE_REPETITIONS=3
```

Workflow:

1) check four U280 artifacts;
2) prepare the six-dataset/four-model inputs;
3) run TempGNN, MATG, ViTeGNN, and RTGA three times;
4) write raw measurement CSVs; and
5) aggregate mean latency.

TABLE II
Expected U280 mean latency.

Implementation	Latency (ms)
TempGNN	1.296553
MATG	9.966618
ViTeGNN	2.869752
RTGA	4.192824

A successful run exits with status zero, records all four implementations, and writes fresh measurements under `results/reviewer_u280_runs/<run-id>/`. TempGNN should retain the lowest mean latency and remain close to 1.3 ms.

TABLE III
Expected A1 review time.

Setup	Execution	Analysis
10–20 min	30–90 min	<5 min

Rebuilding xclbins is optional and is not part of the direct review workflow.

V. Generate Figures from result.csv

`A2` regenerates the paper-result plots associated with `C1–C4`.

```
python3 -m scripts.reproduce_paper_figures
```

The command reads `results/result.csv`, validates and groups the rows, and generates per-figure CSV files, matching SVG figures, and a manifest. Expected output is nine CSV/SVG pairs for Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14 under `results/paper_reproduction/`. The manifest must identify `results/result.csv` as the input.

TABLE IV
Expected A2 review time.

Setup	Execution	Analysis
<5 min	<1 min	2–5 min

VI. Baseline Status

- MATG: independently reproduced from the IPDPS 2022 paper and its partial public HLS headers; includes a distinct source, runner, and U280 xclbin.
- ViTeGNN: independently reproduced from the TPDS 2025 paper; includes a distinct source, runner, and U280 xclbin.
- RTGA: independently reproduced from the DAC 2024 paper; includes a distinct source, runner, and U280 xclbin.
- Cascade and TGLite-CPU: retained only as packaged paper comparison records; no fresh execution of those external stacks is claimed.
- TempGNN: includes HLS kernels, testbenches, Vitis build scripts, XRT host code, U280 board logs, and its own U280 xclbin.

The three baselines are independent, paper-based forward-path reproductions, not the authors' complete original stacks.

VII. Metric Meaning

`DDTC` is dependency-driven TDP construction. `OATS` is overlap-aware TDP synchronization.

`packet_reuse_factor` measures how many per-target packet materializations collapse into unique PHLE packets. `memory_bytes` is packet state/metadata traffic. `cycles` is the kernel's cycle proxy used by C-sim and exported fixture stats. HLS reports provide C/RTL latency after synthesis/cosim.

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